Personalised Offer Engine: Hackathon 2

Punjab Jobs AI Bootcamp · J-PAL South Asia · Individual project

The problem

E-commerce platforms send offers to millions of customers, but not every offer is worth sending. A discount that wins back a customer who is about to leave is valuable; the same discount sent to someone who would have bought anyway is lost margin. Platforms use what they know about each customer, such as purchases, browsing, cart activity and reward points, to decide who receives which offer, and when.

Your task

Build a system that decides which offer each customer of an e-commerce platform should receive, and delivers those offers to your classmates through a Telegram bot. There is no store to build. You will need realistic customer data, so generating it is part of the work. Rules you design set the limits for every offer; an AI agent may choose and personalise offers in real time within those limits, and every AI choice must be checked against your rules before it is sent.

That is the whole brief. Figuring out what the system needs to do is part of the assignment: what data a real platform collects, how customers differ, which offer suits whom, what limits a business would set, and how you would know whether your offers work. Start with something small that works, test it on a classmate, learn what is missing, and build from there.

How you work

- Use Claude Code as your main tool for planning, writing, testing and debugging.
- Commit small working steps to Git.
- Use only synthetic customer data, and keep your bot token and API keys out of your repository.
- Share your bot's link early: a Telegram bot can only message people who have started it.

How winners are chosen

- **Core functionality:** offers stay within the limits your rules set, and reach classmates through the bot.
- **Data realism:** the synthetic data behaves like real customer data.
- **Usability:** a marketing team could understand and adjust the rules without help.
- **Design:** offer messages are clear and appealing, and the results are easy to read.
- **Advanced functionality:** it handles harder or adjacent cases well.
- **Relevance:** the offers make business sense for a real platform.

Timeline

When	What
During class (120 min)	Start building your system.
Tuesday 27 October, 12:00 noon	Submit your work.
Next session	The winners present their solutions.

Submission

1. **GitHub repository** with source code, a README covering setup and usage, and your `CLAUDE.md`.
2. **Telegram bot username**, with the bot running until judging is complete.